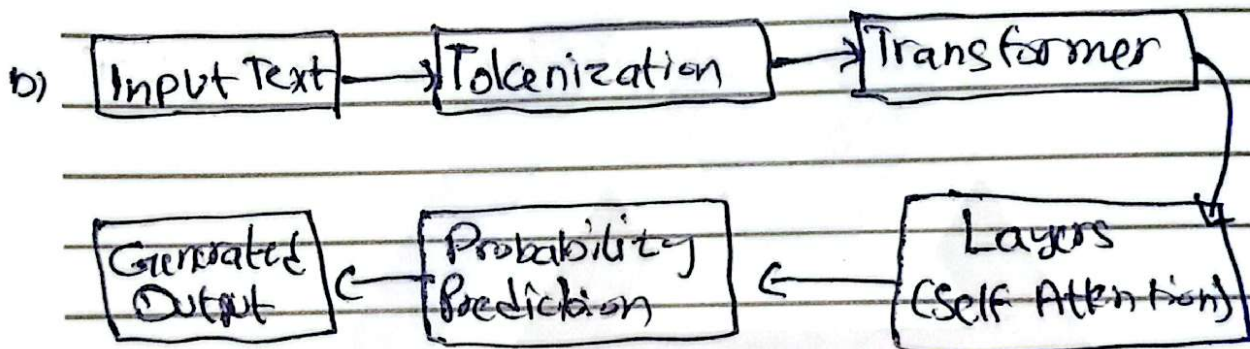




Question 1:

Selected system: ChatGPT.

- a) ChatGPT is designed to understand and generate human-like text. It is based on Transformer architecture and large scale language modeling. Its foundation lies in deep learning, probability prediction and reinforcement learning from human feedback. It reflects AI's goal of enabling machines to process natural language intelligently.



- Tokenizer: Converts text into numerical tokens.
- Transformer layer: Capture context using self attention.
- Prediction Layer: Predict next word probabilities.
- Output Generator: Produces coherent responses.

Each layer contributes to contextual understanding and meaningful text generation.

c) The system includes large training datasets, neural network layers and optimization algorithms. Self-attention allows it to understand relationships between words. Fine tuning with human feedback improves alignment and response quality.

d) ChatGPT has transformed education, business automation and content creation. It improves productivity and access to information worldwide. However, it represents narrow intelligence, as it performs knowledge tasks but lacks true reasoning or consciousness.

e) Technically, it may generate incorrect or biased information. Ethically, it raises concerns about misinformation and job displacement. Practically, it lacks real understanding and relies on statistical prediction.

f) Future Improving factual accuracy and reducing bias are key goals. Better reasoning, multimodal learning and ethical safeguards can enhance reliability. Future versions should focus on transparency and responsible AI development.

Question 2:

- a) One major AI technology in cyber security is AI-based Intrusion Detection System (IDS). Modern systems use Machine Learning and Deep Learning to detect unusual network behaviour.

For example,

Solution developed by companies like Parktrace use AI to identify cyber threats in real time.

- b) AI based IDS collects network traffic data and learns normal behavior patterns. Using anomaly detection and classification algorithms, it identifies suspicious activities. Applications include malware detection, phishing prevention, insider threat detection and real time monitoring of enterprise networks.

- Advantages include faster threat detection, reduced human workload and real time response.
- Challenges include false positives, high implementation cost and dependency on quality data.
- Ethical issues involve data privacy, surveillance concerns and possible misuse of AI for cyber attacks.

d) Future systems should reduce false alarms using better training models.

Integration with threat intelligence platforms can improve adaptability.

Development explainable AI in cybersecurity will increase trust and transparency.

Question 3:

a) PEAS Description.

1. AI Based Resume Screening System

- Performance Measure:

Accuracy of candidate selection, reduced hiring time, fairness in shortlisting.

- Environment:

Online job portals, HR databases, resumes in digital format.

- Actuators:

Shortlist candidates, rank applications, send notifications.

- Sensors:

Resume text, applicant data, keywords jobs descriptions.

2. Flood Detection AI :

• Performance Measure :

Early warning accuracy, reduced false alarms, timely alerts.

• Environment :

Rivers, weather systems, satellite monitoring areas.

- Actuators :

Send alerts, activate warning systems, notify authorities.

• Sensors :

Rainfall data, water levels, satellite images, weather forecasts.

b) Environment characteristics :

1) Resume Screening System :

• Fully Observable :

The agent has access to all data workers' uploaded username file at once.

• Deterministics :

Given the same resume & the same algorithm, the output score will always be the same.

• Episodic:

Each resume is screened independently of the one that came before it.

• Static:

The resume data does not change while the agent is processing it.

• Discrete:

Operates on distinct sets of data & predefined ranking scores.

• Single Agent:

Usually operates as a standalone scoring logic against a database.

2. Flood Detection AI:

• Partially observable:

Many underground or atmospheric factors affecting floods are not fully captured by sensors.

Stochastic:

Weather and fluid dynamics involve random variables that cannot be predicted with 100% certainty.

- Sequential :
Current water levels & rainfall directly affect the probability of future flooding.
- Dynamic :
The ~~river~~ river levels and weather conditions change continuously while the agent.
- Continuous :
Water levels, time and flow rates are continuous physical quantities.
- Multi Agents :
Often interact with other agent (weather station, dam controllers, or city traffic AI).

Question 4:

1. Implicit Agent:

The implicit agent in this scenario is the automated retrieval system (robot) operating inside the university.

2. Formulating as a State Space Search Problem.

We model this as a single agent path finding problem in a grid environment.

• State Space

A state can be represented as

$$\text{State} = (x, y, \text{carrying})$$

- (x, y) → current position of the robot in the grid.
- carrying → Boolean value (True or False) indicating whether the robot is carrying the book.

The state space consists of:

- All valid grid locations (excluding blocked aisles)
- For each location, whether the robot is carrying the book or not.

• Initial State: Initial State = $(x_{\text{start}}, y_{\text{start}}, \text{False})$,

where:

- $(x_{\text{start}}, y_{\text{start}})$ is the fixed starting location.
- $\text{Carrying} = \text{False}$ (robot is not carrying the book).

• Actions:

Possible actions at each state

1. Move Up
2. Move Down
3. Move Left
4. Move Right

5. Pickup.

Constraints:

- Movement only allowed if the next cell is within grid bounds and not blocked.
- Pickup only allowed if robot is at the correct shelf location and not already carrying the book.

• Transition Model:

The transition model defines how actions change the states.

1. Move Action:

$$(x, y, \text{carrying}) \rightarrow (x', y', \text{carrying})$$

2. Pickup Action:

$$(x_{\text{book}}, y_{\text{book}}, \text{false}) \rightarrow (x_{\text{book}}, y_{\text{book}}, \text{true})$$

• Goal Test:

The goal is achieved when:

$$(x, y) = (x_{\text{book}}, y_{\text{book}}) \text{ AND } \text{carrying} = \text{True}$$

Meaning:

- The robot is at the shelf containing the requested book.
- The book has been pickup.

Path Cost:

Each movement step has a cost (e.g., 1 unit of time)

Path Cost = $\sum$ movement steps ,

- Picking up the book may have negligible or fixed cost

- The objective is to minimize total movement cost

Solution:

A solution is :

A sequence of actions that moves the robot from the initial state to the goal state with minimum cost

- Right $\rightarrow$ Right $\rightarrow$ Up $\rightarrow$ Up $\rightarrow$ Left $\rightarrow$ Pickup ,

The optimal solution is the shortest valid path from start location to book shelf avoiding blocked aisles

PREMIUM CHEMICALS

Question 5:

1. The drone is using Iterative Deepening Search (IDS). It performs Depth-limited search up to a fixed depth. If the exit is not found, it increases the depth limit and restarts from the beginning. This continues until the exit is discovered.

2. • Limited Memory:

IDS uses memory like depth-first search which stores only current path. It does not require storing all nodes like Breadth-first search.

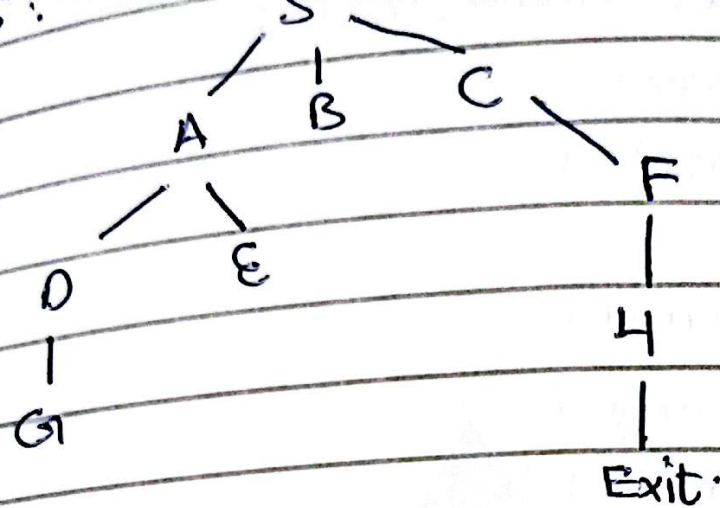
• Unknown Depth of Exit:

Since the exit depth is unknown, IDS gradually increases the limit. This ensures the goal will eventually be found.

• Very Deep Dead End Tunnels

Depth limits prevent the drone from going infinitely deep into dead ends. It avoids getting stuck in very long useless paths.

3. Start S:



How does DFS Explores:

- First iteration: depth limit = 0 $\rightarrow$ check S.
- Second = depth limit = 1 $\rightarrow$ explores A, B, C.
- Third = depth limit = 2 $\rightarrow$ explores D, E, F.
- Fourth = depth limit = 3 $\rightarrow$ explores G, H.
- Fifth = depth limit = 4 $\rightarrow$ finds Exit.

It shows repeated shallow-to-deep exploration until the goal is found.



3. ~~W~~ DFS may go very deep into dead ends before finding the goal.

- If depth limit is too small, DFS fails if too large it wastes time.

- IDS gradually increases depth, guaranteeing finding the shallowest goal.

- It uses low memory like DFS but has completeness like BFS.

Therefore, Iterative Deepening Search is more reliable and efficient for unknown-depth mazes.